M. Kifayath Chowdhury, Ph.D.

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RESEARCH INTERESTS

River morphology and morphodynamics; dynamics of river bifurcations; climate impact and adaptation; sediment transport; environmental fluid mechanics.

EDUCATION

Delft University of Technology, Delft, Netherlands
Ph.D. in Hydraulic Engineering *June 2020-September 2026*
Thesis: *An Engineered River System with Bifurcations under a Changing Climate*

Louisiana State University, Baton Rouge, USA
M.S. in Oceanography & Coastal Sciences *August 2018-May 2020*
Thesis: *Effect of Channelized and Unchannelized Lateral Outflow on Three-Dimensional Flow Structure and Sediment Transport Mechanisms in a River Delta*

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh
B.Sc. in Naval Architecture and Marine Engineering *May 2012-February 2017*
Thesis: *Analysis of Potential Flow around a 3D Hydrofoil Using Vortex Lattice Method*

RESEARCH EXPERIENCE

Doctoral Researcher *June 2020 -September 2026*
Delft University of Technology (Supervisors: Dr. Astrid Blom and Dr. Ralph Schielen)

- Analyzed historic (100 years) hydraulic and morphological dataset from **the Rhine River** system to assess the system's hydro-morphological development
- Developed a one-dimensional hydro-morphological model of the Upper Rhine Delta to assess the future impact of **climate change on river functions**
- Evaluated the influence of climate change on the River Rhine within the Rivers2Morrow project, contributing to collaborative research with Rijkswaterstaat for Room for the River 2.0.
- Mentored and supervised three B.Sc. and two M.Sc. students on their thesis projects, providing guidance on research methodologies, data analysis techniques, and numerical modeling approaches

Graduate Research Assistant *August 2018 -May 2020*
Louisiana State University (Supervisor: Dr. Matthew Hiatt)

- Performed field measurement of velocity and water discharge at Wax Lake Delta, LA, USA
- Analyzed 3D flow data to understand the flow and sediment connectivity in river deltas
- Geo-spatial Data Analysis (NDVI Case Study): Leveraged remote sensing data to analyze vegetation health, utilizing GIS tools to process satellite imagery for planning insights.

PUBLICATIONS

Peer Reviewed

(4) **Chowdhury, M. K.**, Blom, A., Ylla Arbos, C., & Schielen, R. M. J., (2026). Climate-Change Projections of Flow Distribution Across the Rhine Delta. Journal of Geophysical Research: Earth Surface, 131(3). https://doi.org/10.1029/2025JF008705

(3) Blom, A., Ylla Arbos, C., **Chowdhury, M. K.**, Doelman , A., Rietkerk , M., & Schielen, R. M. J., (2024). Indications of Ongoing Noise-Tipping of a Bifurcating River System. Geophysical Research Letters, 51(22). https://doi.org/10.1029/2024GL111846

(2) **Chowdhury, M. K**, Blom, A., Ylla Arbós, C., Verbeek, M. C., Schropp, M. H. I., & Schielen, R. M. J. (2023). Semi-centennial Response of a Bifurcation Region in an Engineered River to Peak Flows and Human Interventions. Water Resources Research, 59(4). https://doi.org/10.1029/2022WR032741

(1) **Chowdhury, M. K**, Konsoer, K. M., & Hiatt, M. (2022). Effect of Lateral Outflow on Three-Dimensional Flow Structure in a River Delta. Water Resources Research, 58(10), 1-20. https://doi.org/10.1029/2021WR031346

PUBLISHED DATASETS (OPEN ACCESS)

(3) **Chowdhury, M. K.**, Ylla Arbós, C., & Schielen, R. M. J. (2025). Schematized Model of the Lower Rhine River and its Branches in SOBEK RE. 4TU.ResearchData. https://doi.org/10.4121/eb78267a-137b-4f61-bb7e-6549915a24c7 (related publication: JGR 2026).

(2) **Chowdhury, M. K.**, Blom, A., Ylla Arbós, C., Verbeek, M. C., Schropp, M. H. I., & Schielen, R. M. J. (2023). Bed elevation, bed surface grain size (D50 & D90), and water discharge Waal, Pannerden Channel, Nederrijn, & IJssel, 19282020. 4TU.ResearchData. https://doi.org/10.4121/19650873 (related publication: WRR 2023)

(1) **Chowdhury, M. K.** (2021). Effect of Lateral Outflow on Flow Structure in a River Delta. Figshare. https://doi.org/10.6084/m9.figshare.15094149 (related publication: WRR 2022)

SKILLS

Programming and Data

- Python (pandas, geopandas, xarray, rasterio, matplotlib)
- MATLAB
- Timeseries analysis for long hydromorphological records; data cleaning, uncertainty analysis

Geospatial Analysis and Remote Sensing

QGIS, ERDAS IMAGINE to integrate of remote sensing products with insitu/river records

Modelling

1D hydro-morphodynamic modeling (SOBEK RE, Delft3D)

Instruments

ADCP, RTK-GPS, Q-boat operations

RELEVANT PROJECTS

LIVING-LAB RHINE (LILAR) - Visiting Researcher *2022*
Participated in cross-border research initiative to learn advanced hydraulic measurement techniques, enhancing skills in data acquisition and analysis relevant to water system assessment.

A literature study of Hydrocarbon Degrading Micro-organisms in the Water Column of the Sundarbans Mangrove Forest *2019*
As part of a graduate-level course, conducted a literature study looking into different species of hydrocarbon-degrading bacteria found in the water column.

Industrial training at Khulna Shipyard Limited, Khulna, Bangladesh *2016*
A month-long training in shipbuilding stages, from design to manufacturing, provided a foundational understanding of engineering processes in a marine context.

SUPERVISION & MENTORING

Current & Former Thesis Students
Rosie Peeters - Hydraulic and morphological response of a river bifurcation to sea level rise in a 1D model.
Tikhon van Oudenallen - Sediment displacement caused by river interventions
Jelle Gortemaker - The grain size gap in river bed surface sediment: a hydromorphological research
S.G.L. van den Brand - Mitigating the gradual change in discharge partitioning at the Pannerdense Kop bifurcation
Rahil Ahmad - Dynamics of the Karnali River Bifurcation in Nepal

VOLUNTEERING EXPERIENCE

Delft University of Technology

Coordinator, M.Sc. course field trip *2023-2024*
Coordinated the logistics of the field trip for the M.Sc. students in the River Engineering track.

Member of Curriculum Redesign- Rivers Unit *2022-2023*
Worked as a team member, prepared lecture materials in the Department of Hydraulic Engineering.

Member of Local Organizing Committee, NCR Days 2022 *January 2022-April 2022*
Helped with logistics and was in charge of the hybrid module of the yearly conference hosted by TU Delft in 2022.

Louisiana State University

Secretary, Coast and Environment Graduate Organization (CEGO) *May 2019-May 2020*
Worked as secretary of the faculty graduate students' association and organized outreach events.

Sports Coordinator, Bangladeshi Students Association *June 2019-May 2020*
Worked as the coordinator of sports activities and organized events within the student community.

Bangladesh University of Engineering and Technology

Organizer, Model Ship Propulsion Competition-2016, *November 2016*
Managed a team of 20 volunteers to organize a scaled model ship racing competition.

Volunteer and Student Committee Member, MARTEC-2016 *October 2016-December 2016*
Looked after the logistics of organizing the conference and hosted international students visiting Dhaka for the conference.

Sub-editor *July 2012- October 2015*
Zero2infinity, Science Magazine, Dhaka, Bangladesh
Reviewed and edited articles on Astronomy in the local Bengali science magazine

HONORS & AWARDS

Best Presentation Award, NCR Days 2023, Nijmegen, The Netherlands, 2023

Graduate Student Association Travel Award, Louisiana State University, 2019

CSI Graduate Student Travel Award, Louisiana State University, 2019

Faculty of Mechanical Engineering Dean's List, Bangladesh University of Engineering and Technology, 2014-2015

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

American Geophysical Union (AGU) *2019 - Present*
Coastal Studies Institute (LSU-CSI) *2019 - 2020*
The Institution of Engineers, Bangladesh (IEB) *2018 - Present*

INVITED PRESENTATIONS

Future of the Upper Dutch Rhine Bifurcation region under climate change, NCR Summer School, July 2025.

Morphological development of the Upper Dutch Rhine Bifurcation region over the last century, NCR Summer School, July 2023.

The evolving Upper Dutch Rhine Bifurcation region: insights from the last century and lessons for the future, NCR theme day-Discharge Distribution, September 2023.

SELECTED CONFERENCE ABSTRACTS & PRESENTATIONS

$'$ *indicates presenting author*

Chowdhury, M.K., Blom, A.$'$, Ylla Arbós, C., & Schielen, R. M. J., Climate Change Impact on a River Bifurcation Region, 14^{th} Symposium on River, Coastal and Estuarine Morphodynamics Meeting (RCEM), Barcelona, September 2025.

Chowdhury, M.K., Blom, A., Ylla Arbós, C., & Schielen, R. M. J.$'$, Projected climate-change impacts on the flow partitioning in the bifurcation region in the upper Rhine delta, NCR Days 2025, Deltares, Delft, The Netherlands, May 2025.

Chowdhury, M.K., Blom, A., & Schielen, R. M. J.,Response of a river bifurcation region to climate-induced hydrograph changes, River Flow 2024, Liverpool, September 2024.

Chowdhury, M.K.$'$, Blom, A., Ylla Arbós, C., & Schielen, R. M. J., Developing a Model to Study the Climate Change Impact on River Bifurcations in Engineered Rivers, NCR Days 2024, Wageningen University & Research campus, Wageningen, Netherlands, February 2024.

Blom, A.$'$, Schielen, R. M. J., Ylla Arbos, C., Doelman , A., Rietkerk , M., & **Chowdhury, M. K.**, A Possible Sign of Fluvial System Tipping, American Geophysical Union (AGU) Fall Meeting, San Francisco, CA, December 2023.

Chowdhury, M.K.$'$, Blom, A., & Schielen, R. M. J., How do Bifurcations in Engineered Rivers Respond to Sea level Rise, 13^{th} Symposium on River, Coastal and Estuarine Morphodynamics Meeting (RCEM), Urbana, IL, September 2023.

Chowdhury, M.K.$'$, Blom, A., Ylla Arbós, C., Verbeek, M. C., Schropp, M. H. I., & Schielen, R. M. J., Response of the Upper Dutch Rhine Bifurcation Region to Peak flows, NCR Days 2023, Nijmegen, Netherlands, April 2023.

Chowdhury, M.K.$'$, Blom, A., Ylla Arbós, C., Verbeek, M. C., Schropp, M. H. I., & Schielen, R. M. J., How Does a River Bifurcation System Respond to Peak Flows? A Case Study of the Upper Dutch Rhine Bifurcation Region, American Geophysical Union (AGU) Fall Meeting, Chicago, IL, December 2022.

Blom, A., Schielen, R. M. J., Ylla Arbos, C., Doelman , A., Rietkerk , M., & **Chowdhury, M. K.**$'$, Evidence of tipping of a fluvial system?, American Geophysical Union (AGU) Fall Meeting, Chicago, IL, December 2022.

Chowdhury, M.K.′, Blom, A., & Schielen, R. M. J., Flow partitioning at a bifurcation in the Upper Dutch Rhine, 11^{th} International Conference on Fluvial Hydraulics, River Flow 2022, Canada, November 2022.

Chowdhury, M.K.′, A. Blom, C. Ylla Arbos, Merel C. Verbeek, Max H. I. Schropp, & R.M.J. Schielen, Channel Bed Erosion Characteristics in the Upper Dutch Rhine Bifurcation Region, NCR DAYS 2022: Anthropogenic Rivers, Delft University of Technology, The Netherlands, April 2022.

Chowdhury, M.K.′, A. Blom, & R. Schielen, (2021). Bifurcation Morphodynamics in an Engineered River, River, Coastal and Estuarine Morphodynamics Meeting (RCEM), November 2021. **Chowdhury, M.K.′**, A. Blom, & R.M.J. Schielen, (2021). Flow Structure at the Pannerdense Kop Bifurcation, NCR DAYS 2021: Rivers in an uncertain future, University of Twente, The Netherlands.

Chowdhury, M.K.′, K.M. Konsoer, and M. Hiatt, Effects of backwater and lateral outflow on three-dimensional flow structure in a river-dominated delta, American Geophysical Union (AGU) Fall Meeting, San Francisco, CA, December 2019.